In vitro binding capacities of three dietary fibers and their mixture for four toxic elements, cholesterol, and bile acid.

Water-soluble dietary fibers from apple peels and water-insoluble dietary fibers from wheat bran and soybean-seed hull were used to evaluate their binding capacities for four toxic elements (Pb, Hg, Cd, and As), lard, cholesterol, and bile acids. The water-soluble dietary fibers showed a higher binding capacity for three toxic cations, cholesterol, and sodium cholate; and a lower binding capacity for lard, compared to the water-insoluble ones. A mixture of the dietary fibers from all samples - apple peels, wheat bran, and soybean-seed hull - in the ratio 2:4:4 (w/w) significantly increased the binding capacity of water-insoluble dietary fibers for the three toxic cations, cholesterol, and sodium cholate; moreover, the mixture could lower the concentrations of Pb(2+) and Cd(+) in the tested solutions to levels lower than those occurring in rice and vegetables grown in polluted soils. However, all the tested fibers showed a low binding capacity for the toxic anion, AsO(3)(3-). Copyright © 2010. Published by Elsevier B.V.